

Course Code: EET305

Course Name: SIGNALS AND SYSTEMS

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Check the system given by $y(t)\frac{d^2y}{dt^2} + 3t\frac{dy}{dt} + y(t) = x(t)$ for i) causality | 3 |
| | ii) linearity and iii) time invariance | |
| 2 | Sketch the signal $u(t) - u(t - 2)$, where $u(t)$ is a unit step signal. | 3 |
| 3 | Explain the Dirichlet's conditions for existence of Fourier Transform. | 3 |
| 4 | State and prove Parseval's theorem applied to Fourier series. | 3 |
| 5 | Explain the concept of positive real functions | 3 |
| 6 | Derive the unit step response of a first order system | 3 |
| 7 | Explain a ZOH circuit and derive its transfer function | 3 |
| 8 | Find the Nyquist rate and Nyquist interval for the signal, $x(t) = \sin 200\pi t$ | 3 |
| 9 | State and prove the time reversal property of discrete time Fourier series. | 3 |
| 10 | Find the Fourier transform of the discrete signal, $x[n] = a^n u[n]$ | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|--|---|
| 11 | a) Explain the different types and properties of non- linear systems | 7 |
| | b) Find the output of a system with impulse response $h(t) = e^{-3t}u(t)$ and the input signal, $x(t) = u(t - 3) - u(t - 5)$ | 7 |
| 12 | a) For the signal shown in Fig. 1, find the signals, | 7 |
| | i) $x(t - 2)$ | |
| | ii) $x(-2t + 3)$ | |

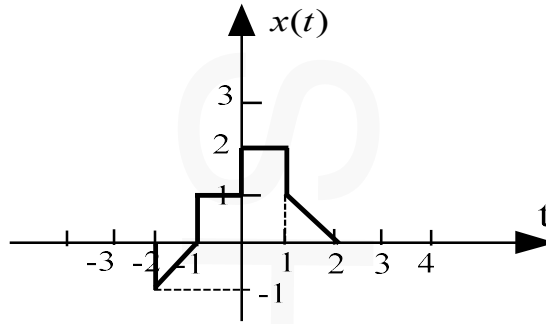


Fig. 1

- b) Differentiate energy and power signals. Determine the energy and power of the signal, $x(t) = (1 + e^{-5t})u(t)$ 7

Module -2

- 13 a) Obtain the trigonometric Fourier series expression of a half-rectified sine wave of amplitude, A. 7
- b) Find the Fourier transform of the following function: 7

$$x(t) = \cos \omega_0 t u(t)$$

- 14 a) Write the differential equations governing the behaviour of the mechanical system shown in Fig. 2. Also obtain an analogous electrical circuit based on force- current analogy. 8

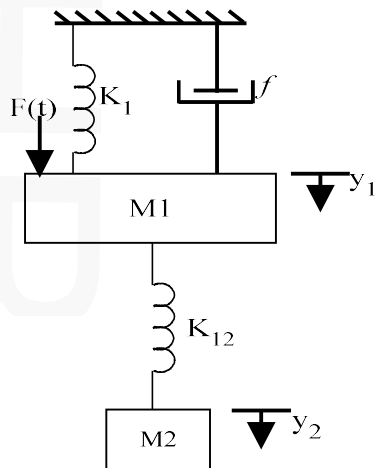


Fig. 2

- b) The differential equation of a causal, continuous- time LTI system is given by 6
- $$\ddot{y}(t) + \dot{y}(t) - 2y(t) = x(t).$$
- a) Find the transfer function $H(s)$ of the system.
- b) Find the impulse response $h(t)$ of this system.

Module -3

- 15 a) Obtain the overall transfer function, $C(s)/R(s)$ of the signal flow graph shown in Fig. 3. 14

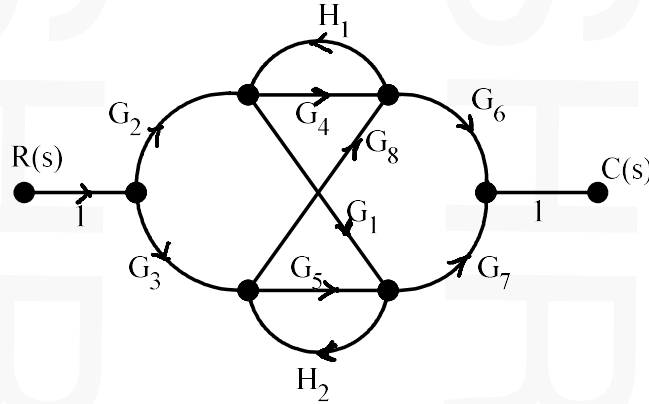


Fig. 3

- 16 a) State the necessary and sufficient conditions to be satisfied for stability of an LTI system. Prove that the necessary condition only has to be satisfied for first and second order systems. 7
- b) Analyse the stability of the given LTI system, whose characteristic equation is given by $Q(s) = s^5 + 2s^4 + 2s^3 + 4s^2 + 11s + 10$, using Routh's criterion. 7

Module -4

- 17 a) Find the inverse z- transform of $X(z) = \frac{1 - \frac{1}{3}z^{-1}}{(1 - z^{-1})(1 + 2z^{-1})}$ ROC: $|z| > 2$ 7
- b) Determine the convolution of the signals: 7
- $$x(n) = \cos \pi n u(n); \quad h(n) = \left(\frac{1}{2}\right)^n u(n)$$
- 18 a) State and prove the associative property of discrete convolution. 7
- b) Find the z- transform of $x[n] = n[u[n] - u[n-8]]$ 7

Module -5

- 19 a) Find the impulse response and frequency response of the system described by the difference equation, $y[n] - \left(\frac{1}{6}\right)y[n-1] + \left(\frac{1}{6}\right)y[n-2] = x[n-1]$ 14

- 20 a) Check the stability of the system whose characteristic equation is given by , 6
 $z^3 + 0.2z^2 - 0.25z + 0.05 = 0$, using Jury's stability criterion.

- b) The system function of a discrete time system is 8

$$H[z] = \frac{(1+z^{-1})^4}{\left(1-z^{-1}+\frac{7}{8}z^{-2}\right)\left(1+2z^{-1}+\frac{3}{4}z^{-2}\right)}$$

Realize this system using s cascade of second order system in direct form- II.
